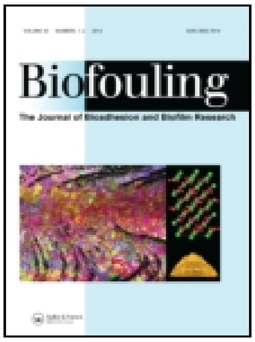

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Kinetics of biofilm formation and desiccation survival of *Listeria monocytogenes* in single and dual species biofilms with *Pseudomonas fluorescens*, *Serratia proteamaculans* or *Shewanella baltica* on food-grade stainless steel surfaces

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This study investigated the dynamics of static biofilm formation (100% RH, 15 °C, 48–72 h) and desiccation survival (43% RH, 15 °C, 21 days) of *Listeria monocytogenes*, in dual species biofilms with the common spoilage bacteria, *Pseudomonas fluorescens*, *Serratia proteamaculans* and *Shewanella baltica*, on the surface of food grade stainless steel. The Gram-negative bacteria reduced the maximum biofilm population of *L. monocytogenes* in dual species biofilms and increased its inactivation during desiccation. However, due to the higher desiccation resistance of *Listeria* relative to *P. fluorescens* and *S. baltica*, the pathogen survived in greater final numbers. In contrast, *S. proteamaculans* outcompeted the pathogen during the biofilm formation and exhibited similar desiccation survival, causing the $N_{21\ \text{days}}$ of *Serratia* to be *ca* 3 $\text{Log}_{10}(\text{CFU cm}^{-2})$ greater than that of *Listeria* in the dual species biofilm. Microscopy revealed biofilm morphologies with variable amounts of exopolymeric substance and the presence of separate microcolonies. Under these simulated food plant conditions, the fate of *L. monocytogenes* during formation of mixed biofilms and desiccation depended on the implicit characteristics of the co-cultured bacterium.

Keywords: *Listeria monocytogenes*; desiccation tolerance; mixed species biofilms; spoilage bacteria; Jameson effect; scanning electron microscopy

Introduction

Listeria monocytogenes has been widely recognized as a food-borne pathogen after contaminated coleslaw caused an outbreak of listeriosis in Halifax, Nova Scotia in 1981 (Schlech III 2000). In spite of intense research efforts large food-borne listeriosis outbreaks continue to be reported including the 2008 luncheon meat outbreak in Canada with 56 cases of human illness and 22 deaths (Weatherill 2009), and the 2011 cantaloupe outbreak in the USA with 146 human cases and 30 deaths (CDC 2011). Clearly, better intervention strategies (as in, for example, cleaning/sanitation chemistry and management, and equipment design) are urgently needed.

L. monocytogenes can persist in food processing plants leading to the same genotype being re-isolated for years despite cleaning efforts and periods with inactivity (Wulff et al. 2006; Keto-Timonen et al. 2007). Its survival on food contact surfaces increases the risk of foods being recontaminated during the steps of processing and packaging (Midelet et al. 2006; Rodriguez & McLandsborough 2007; Keskinen et al. 2008). As persistence of *L. monocytogenes* in food plants depends on the relationship between introduction, survival, growth and removal (Carpentier & Cerf 2011), it is important to understand the impact that biofilm formation and tolerance to stresses (eg heat, salt and low-relative humidity [RH]) have on its survival and growth.

The formation of biofilm by *L. monocytogenes* diminishes the efficiency of cleaning and sanitation efforts. The development of *L. monocytogenes* biofilms depends on factors such as serotype, nutrients, temperature and flow (Herald & Zottola 1988; Chae & Schraft 2000; Kalmokoff et al. 2001; Pan et al. 2010; Nilsson et al. 2011). Rieu et al. (2008) found that *L. monocytogenes* under static conditions simulating a food contact surface (ie absence of flow) formed an unstructured biofilm consisting of a few layers of cells rather than the structured biofilm consisting of a knitted network produced under flow conditions. Analyses of the exopolymeric substances (EPS) of the biofilm of the bacterium have revealed the presence of carbohydrates (Hefford et al. 2005; Chae et al. 2006), poly-γ-glutamate (Nwaiwu et al. 2010), extracellular DNA (eDNA, Harmsen et al. 2010) and proteins (Longhi et al. 2008; Abee et al. 2011).

In natural environments as well as in food processing facilities, *L. monocytogenes* is likely found in mixed species biofilms. While some food-spoilage bacteria such as *Pseudomonas putida*, *P. fragi*, *P. fluorescens* and *Flavobacterium* have been reported to enhance the adhesion, colonization and the formation of biofilm by

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L. monocytogenes (Bremer et al. 2001; Carpentier & Chassaing 2004), several lactic acid bacteria (Guerrieri et al. 2009; Winkelstroter et al. 2011) and *Staphylococcus sciuri* (Leriche & Carpentier 2000) inhibited *L. monocytogenes* biofilms.

Fluctuating RH in food plants and extended periods of inactivity during plant closures will lead to desiccation of microorganisms lodging on processing equipment. Strains of *L. monocytogenes* survived desiccation for three months in a simulated dry food processing environment (Vogel et al. 2010). Moreover, the presence of organic (food) soils, biofilm and salt has been shown to enhance the desiccation survival of the bacterium (Takahashi et al. 2011; Truelstrup Hansen & Vogel 2011; Hingston et al. 2013). The tolerance of bacteria to water stress is related to the cellular genetics and environmental conditions including the presence of hydroscopic EPS (Potts 1994). At this time, the desiccation survival of *L. monocytogenes* in a mixed biofilm has only been investigated in the study by Bremer et al. (2001), where the presence of *Flavobacterium* spp. significantly enhanced listerial survival. Also, investigations into the kinetics of formation of dual species biofilm with *L. monocytogenes* are limited.

The aim of this study was to investigate the dynamics of formation of biofilm and desiccation inactivation of *L. monocytogenes* in static dual species biofilms with three common food-spoilage organisms, *P. fluorescens*, *S. proteamaculans* and *S. baltica*, on food-grade stainless steel (SS). The growth of biofilm (15 °C, 100% RH for 48–72 h) and desiccation survival kinetics (15 °C, 43% RH for 21 days) of each bacterium in single and dual species biofilms were quantified and described using non-linear regression models. Also, the biofilm structure was characterized using scanning electron microscopy (SEM) and epi-fluorescence microscopy.

Materials and methods

The strains of bacteria, preparation of stocks and inocula

Four different strains of bacteria were used in this study: *S. baltica* A2 (isolated from spoiled cod and previously identified as *Shewanella putrefaciens* A2, Ravn Jørgensen et al. 1988), *S. proteamaculans* 2R4 (from spoiled cold-smoked salmon and originally identified as *Serratia liquefaciens* 2R4, Truelstrup Hansen 1995), *P. fluorescens* ATCC 13525 (ATCC®, Manassas, VA, USA) and *L. monocytogenes* 568 (serotype 1/2a, Kalmokoff et al. 2001). Each bacterial strain was maintained on brain heart infusion agar (BHIA) (Difco, BD Canada, Oakville, ON) and cultures renewed monthly from frozen stock cultures stored at −80 °C in BHI with 20% glycerol (Sigma, Oakville, ON).

Strains A2 and 2R4 were re-named following amplification by PCR of their 16S rRNA genes using the universal primers F44 and R1543 as described by Abnous et al. (2009), and Sanger sequencing of the PCR products. The sequences obtained (1.5 kbp) were aligned using the BLASTn software (blast.ncbi.nlm.nih.gov) and based on the *E*-values assigned an identity.

In order to harvest early stationary-phase cells for use in the experiments, the growth kinetics of each bacterial strain were determined at 15 °C. Each strain was inoculated into Tryptic Soy Broth (TSB) (Bacto, BD Canada, Oakville, ON, USA) supplemented with 1% (w v^{-1}) glucose (TSB-glu) and incubated at 15 °C for 72 h. Bacterial counts were obtained by spread plating on BHIA (30 °C, 48 h) at specific time intervals. The growth curves for planktonic growth were constructed, and early stationary phase was determined to occur after 24 h for *P. fluorescens* and *S. proteamaculans* and after 48 h for *S. baltica* and *L. monocytogenes*, respectively (data not shown here).

Formation of single and dual species biofilms on stainless steel surfaces

A single colony from each bacterial strain was inoculated into 5 ml of TSB-glu and incubated at 15 °C for 24 (*P. fluorescens* and *S. proteamaculans*) or 48 h (*L. monocytogenes* and *S. baltica*) to obtain early stationary phase cells. Following growth at 15 °C, the bacterial cells were pelleted (3396 × *g*, 15 min, 4 °C) and resuspended in fresh TSB-glu. The concentration of each cellular suspension was adjusted to ≈10^9 CFU ml^{-1} ($ABS_{450\ nm}$ = 1.0) and serially diluted in fresh TSB-glu to 10^6 CFU ml^{-1}.

Stainless steel (SS, 316, type 4 finish) coupons with the dimensions of 0.5 × 0.5 cm were degreased by boiling in 1% (w v^{-1}) sodium dodecyl sulphate (Bio-Rad Laboratories Canada, Mississauga, ON, Canada) for 10 min. The coupons were then rinsed three times with distilled water (dH_2O) and sonicated (Elmo Ultrasonic bath, 50/60 Hz, Fisher Scientific) in 15% (v v^{-1}) Decon solution (CiDecon® concentrated phenolic disinfectant, Decon Labs, Fisher Scientific) for 1 h. Following the second rinse in dH_2O, the coupons were passivated in acetic acid (10% v v^{-1}, Fisher Scientific) for 10 min, rinsed again with dH_2O and autoclaved for 15 min at 121 °C. The sterile coupons were then dried and finally stored in 95% ethanol until future use.

For formation of single species biofilms, 10 µl of each culture were deposited onto the surface of cooled flame-sterilized SS coupons to yield an initial concentration of 10^3 CFU cm^{-2}. The inoculated coupons were transferred to a desiccation chamber (Scienceware Desiccator Cabinet, Fisher Scientific, Ottawa, ON, Canada), where the RH had been adjusted to 100% by placing three Petri dishes filled with distilled water (dH_2O) in the bottom of the chamber, and incubated at 15 °C for 48–72 h to allow for the formation of biofilm.

Dual species biofilms were formed by placing 5 µl of the Gram-negative strain (*P. fluorescens*, *S. proteamaculans* or *S. baltica*) suspension together with 5 µl of the *L. monocytogenes* suspension on SS coupons to yield a total initial concentration of 10^3 CFU cm^{-2}. Biofilms were formed as described above (15 °C, 100% RH, 72 h).

Enumeration of bacterial growth during formation of single and dual species biofilms

The growth of the bacteria in the single and dual species biofilms was determined by spread plating on suitable general (BHIA) or selective agars (Oxford, *Pseudomonas* C–F–C, VRBG, Iron agar + Pen G, see below) to enumerate the populations developing on the SS coupons over the course of the formation of biofilm, with sample collection performed after 0, 12, 24, 36, 48, 60 and 72 h.

For each single species of biofilm, three coupons were randomly selected at each sampling time, gently rinsed three times by dipping the coupons in physiological peptone saline (PPS, 0.1% peptone, 0.85% NaCl) to remove the loosely attached cells from the surface, and then placed in microcentrifuge tubes containing 0.99 ml of PPS. The tubes were vortexed for 30 s, sonicated for 4 min, vortexed again for 30 s, serially diluted in PPS and spread plated onto both BHIA and the appropriate selective medium. *Listeria* selective agar base (Oxford formulation, CM0856, Oxoid, Nepean, ON, Canada) with *Listeria* selective supplement (SR0140, Oxoid) was used for *L. monocytogenes*. Specific counts of *P. fluorescens* were obtained on *Pseudomonas* agar base (CM0559, Oxoid) supplemented with *Pseudomonas* C–F–C (SR0103, Oxoid), while *S. proteamaculans* was enumerated on violet red bile glucose (VRBG) agar (CM1082, Oxoid). *S. baltica* was counted as black (H_2S-producing) colonies on Iron agar (Gram et al. 1987) supplemented with 0.6 µg ml^{-1} of penicillin G (Pen G potassium salt 10 MU, Sigma-Aldrich, Oakville, ON, Canada). The entire experiment was performed twice (biologically independent replicates) for each bacterial biofilm using triplicate samples (n = 6). Counts (CFU cm^{-2}) were log-transformed and growth curves were constructed.

For dual species biofilms (n = 6), the same protocol was utilized except samples were only spread plated onto the appropriate selective agars, ie Oxford agar for *L. monocytogenes* and the matching selective media for the Gram-negative bacteria. Populations in single and dual species biofilms reached the stationary phase after 48 h, indicating that mature biofilms had formed.

Desiccation survival in single and dual species biofilms on SS surfaces

The single and dual species biofilms were prepared on SS coupons as described above. After incubation (15 °C, 100% RH, 48 h) to allow for formation of mature biofilms, the Petri dishes with coupons were transferred into a desiccation chamber (Mini desiccators, W × D × H: 224 × 200 × 168 mm, Bohlender, Grünsfeld, Germany) equipped with four Petri dishes filled with saturated potassium carbonate (Fisher Scientific) to serve as desiccant (43% RH). The desiccation chamber was then placed at 15 °C. The temperature and RH in the desiccation chamber were continuously monitored using a data logger (TV-4500, Tinytag Canada, Markham, ON) and remained constant throughout the experimental desiccation period of 21 d.

At specific time intervals [−2 (initiation of formation of biofilm), 0 (end of formation of biofilm, beginning of desiccation period), 1, 3, 5, 7, 14, 21 d], three coupons from each biofilm treatment were randomly sampled. Survivors in single species biofilms were enumerated on BHIA while desiccation survivors in the dual species biofilms were enumerated on the appropriate selective agar for each of the two bacterial species as described above. All experiments were repeated in two biologically independent runs (n = 6). The desiccation survival in single species biofilms made of *L. monocytogenes* and *S. proteamaculans* was further investigated using additional sampling times of 2, 4, 6, 9, 11, 13, 15, 17 and 19 d.

Modelling of the growth and formation of single and dual species biofilms on SS surfaces

Populations in single and dual species biofilms consisting of strains of *L. monocytogenes* and the Gram-negative bacterial strains on SS coupons were converted into Log_{10} CFU cm^{-2} and presented as mean ± standard deviation (SD) ($n-1$) for each strain. The growth curves were plotted using SigmaPlot® software for Windows version 11.0 (2008, Systat Software Inc, San Jose, CA, USA). The non-linear Logistic with lag phase growth model was fitted to the data using the Solver add-in for Microsoft® Office Excel® (2007, Microsoft Corporation, Redmond, WA, USA).

The Logistic with lag phase model was parameterized as follows (Dalgaard 2009):

$$\begin{cases} \text{Log}_{10}(N_t) = \text{Log}_{10}(N_0) & t \le \lambda \\ \text{Log}_{10}(N_t) = \text{Log}_{10}\left(\dfrac{N_{\max}}{1+\left[\frac{N_{\max}}{N_0}-1\right]\times\exp\left(-\mu_{\max}\times(t-\lambda)\right)}\right) & t > \lambda \end{cases} \tag{1}$$

where N_{max} is the maximum population (CFU cm^{-2}), μ_{max} represents maximum specific growth rate (h^{-1}), λ denotes lag time (h), N_0 shows initial populations at time 0 (CFU cm^{-2}), t is time (h) and N_t is number of cells at any time (CFU cm^{-2}).

Modelling of bacterial desiccation survival in single and dual species biofilms on SS surfaces

Survivor counts obtained for *L. monocytogenes* and the Gram-negative bacteria during desiccation of single and dual species biofilms on SS coupons were Log_{10}-transformed, normalized by conversion into $\text{Log}_{10}(N_t\ N_0^{-1})$ and presented as mean ± SD for each strain. The survivor curves were plotted using the SigmaPlot software. The non-linear double Weibull inactivation model was fitted to each of the six replicate survivor curves obtained for biofilm treatment using the free curve-fitting tool GInaFiT (version 1.6) developed by Geeraerd et al. (2005) for Microsoft® Office Excel® and available at the KULeuven/BioTec-homepage (http://cit.kuleuven.be/biotec/downloads.php).

The double Weibull model, which assumes the presence of two different subpopulations (1 and 2) with differing resistance to the inactivation treatment, had in preliminary analyses (data not shown) been selected as the model that best described the kinetics of desiccation survival in all types of biofilm and is parameterized as follows (Coroller et al. 2006):

$$\text{Log}_{10}\left(\frac{N_t}{N_0}\right) = \text{Log}_{10}\left[10^{-\left[\left(\frac{t}{\lambda_1}\right)^p+\alpha\right]} + 10^{-\left(\frac{t}{\lambda_2}\right)^p}\right] - \text{Log}_{10}(1+10^{\alpha}) \qquad (2)$$

where $\frac{N_t}{N_0}$ denotes the relative number of total survivors, α (Equation 3) represents the Logit of 'f' (f is the fraction of subpopulation 1 in the total population) which may also be shown as $\text{Log}_{10}(N_{01}\ N_{02}^{-1})$, δ_1 is first decimal reduction time for subpopulation 1 (d), δ_2 shows first decimal reduction time for subpopulation 2 (d), t is time (d) and p denotes the curve shape factor.

$$\alpha = \text{Log}_{10}\left(\frac{f}{1-f}\right) \qquad (3)$$

Statistical analysis

The estimates of model parameters, which had been obtained for each of the six replicates within each biofilm treatment for the Logistic with lag phase growth (ie N_{max}, μ_{max}, N_0 and λ) and double Weibull inactivation (ie α, δ_1, δ_2, and p) models, were compared between biofilm treatments using analysis of variance (ANOVA) with Tukey's Studentized Range (HSD) test at the 5% significance level using SAS software V 9.2. Also, the goodness of the model fits obtained for each biofilm were assessed using three indices (r^2, MSE_{model} and the F-test) as described by Ells et al. (2009).

Values for the initial biofilm populations ($N_{0\ days}$), final desiccation survival numbers ($N_{21\ days}$) and desiccation viability loss (ΔLog_{10} ($N_{21}\ N_0^{-1}$)) were compared among all biofilms using ANOVA with Tukey's Studentized Range (HSD) test ($p < 0.05$).

Microscopic evaluation of adhesion and formation of biofilm

Surface SEM

Two fixation protocols (one hydrous and one anhydrous) were used to prepare the SS coupons harbouring single or dual species biofilms for observation under SEM.

Modified hydrous fixation in cacodylate buffer, aqueous tannic acid and uranyl acetate. The cacodylate buffer fixation technique based on the method of Dekker et al. (1991) was employed where following the formation of biofilm as previously described, the SS coupons were submerged in 0.2 M of cacodylate buffer (pH 7.2) with 2.5% (w v^{-1}) glutaraldehyde and 0.05 M calcium chloride (Fisher Scientific) for 2 h. The coupons were then rinsed (3 × 10 min) in 0.1 M cacodylate buffer (pH 7.2) supplemented with 5% sucrose (Fisher Scientific) and immersed for 4 h in 1% (v v^{-1}) osmium tetroxide solution in cacodylate/sucrose buffer. After the second rinse (3 × 10 min) in the cacodylate/sucrose buffer that was followed by 2 × 5 min rinsing in filter-sterilized dH_2O, the coupons were transferred into 1% (w v^{-1}) aqueous tannic acid (Mallinckrodt Canada, Pointe-Claire, QC, Canada) for 30 min, rinsed (3 × 10 min) in dH_2O and submerged in 2% (w v^{-1}) aqueous uranyl acetate (Taab laboratories, Canton de Gore, QC, Canada) for 30 min. Following 3 × 10 min rinsing in dH_2O, the fixed biofilms were dehydrated in an ascending ethanol (Fisher Scientific) gradient series (35, 50, 70, 90 and 100%, 15 min in each except the last one which was repeated three times) and dried in a HMDS/ethanol (hexamethyldisilazane, Electron Microscopy Science (EMS), Cedarlane, Burlington, ON, Canada) mixture series (25:75, 50:50, 75:25 and 100:0, 15 min for each step except the last one which was repeated twice).

The fixed, dehydrated and chemically dried coupons were air dried for 2 h, mounted onto the aluminum mounts (slotted head, tapered pin and EMS) using carbon adhesive tabs (9 mm diameter, EMS) and sputter coated (Polaron-SC7620 mini sputter coater, Quorum Technologies Ltd Canada, Montréal, QC, Canada) with Au/Pd nanoparticles (SC502-314B gold/palladium sputter target, 0.1 mm thick, Quorum Technologies Ltd). The prepared coated coupons were protected in universal reversible mount holders (EMS) and stored under almost dry conditions (3–4% RH) for future observations by the SEM instrument.

Anhydrous fixation in FC-72 solvent. The protocol developed by Allan-Wojtas et al. (1997) using an anhydrous solvent-based fixation protocol was applied to single and dual species biofilms formed (48 h, 15 °C, 100% RH) on SS coupons as before. Briefly, SS coupons were then transferred into 1% (w v^{-1}) osmium tetroxide crystalline (EMS) dissolved in FC-72 solvent (3 M™ Flourinert™ Electronic Liquid, 3M, London, ON, Canada) and left in the solution for 4 h. After immersion in pure FC-72 solvent for 30 min, the coupons were placed in cacodylate buffer solution (0.2 M) containing glutaraldehyde (2.5% w v^{-1}) and calcium chloride (0.05 M) for 2 h. Subsequently, the coupons were rinsed in dH_2O (2 × 5 min), submerged in 1% (w v^{-1}) aqueous tannic acid (30 min), rinsed again in dH_2O (3 × 10 min), immersed in 2% (w v^{-1}) aqueous uranyl acetate (30 min) and finally rinsed in dH_2O (3 × 10 min). The fixed coupons then were dehydrated, dried, mounted, coated and stored using the same protocols as stated before.

SEM observations of bacterial biofilms formed on the SS coupons. The stubs containing the fixed and coated coupons were mounted on the microscope specimen holder and placed inside the microscope chamber according to the operating instructions. The structure of the biofilms was visualized with the field-emission surface scanning electron microscope (Hitachi S-4700 FE-SEM) using operational conditions of 10 kV of acceleration voltage (V_{acc}), 20 µA of emissions current, 7.5–8.5 mm working distance and the UHR-A lens mode. Micrographs were taken from multiple areas of each coupon (two coupons per biofilm treatment and three random pictures from randomized areas of each coupon at various magnifications).

Epifluorescence microscopy

The Live BacLight™ Bacterial Gram stain kit (L-7005, Life Technologies Inc, Burlington, ON, Canada) and epifluorescence microscopy were used to differentially observe the bacteria within the mixed species biofilms (see Supplementary material for details). (Supplementary material is available *via* a multimedia link on the online article webpage.)

Results

Growth of L. monocytogenes *and the Gram-negative bacteria during formation of single and dual species biofilms on SS coupons*

The population of *P. fluorescens* rose at the highest maximum specific growth rate (μ_{max} = 0.38 h^{-1}) among all single species biofilms, although statistically this was not significantly different (p > 0.05) from the μ_{max} (0.36 h^{-1}) of *L. monocytogenes* (Figure 1 a and b, Table 1). The μ_{max} of *S. proteamaculans* and *S. baltica* in single species biofilms were similar and significantly (p < 0.05) lower than the growth rates obtained for the other two bacteria (Figure 1 c and d, Table 1). *L. monocytogenes* was the only bacterium that showed a lag phase (λ = 19.1 h) in the single species biofilms, causing it to reach N_{max} after 48 h (Figure 1a, Table 1). In contrast, the Gram-negative bacteria showed no lag phase in the single species biofilms and reached their respective N_{max} after ~36 h (Figure 1b–d, Table 1).

The maximum population density (MPD, indicated as N_{max} in Table 1) obtained in the single species biofilms differed significantly (p < 0.05) among the four strains, with *S. baltica* reaching the highest levels of $10^{7.84}$ CFU cm^{-2} followed by *S. proteamaculans* ($10^{7.46}$ CFU cm^{-2}), *P. fluorescens* ($10^{7.25}$ CFU cm^{-2}) and finally *L. monocytogenes* with levels of $10^{6.83}$ CFU cm^{-2} (Table 1, open symbols in Figure 1a–d). N_0 values predicted by the model came close to the measured numbers spotted on the SS coupons of ~10^3 CFU cm^{-2} (Table 1 and Figure 1a–d).

The competitor Gram-negative strains had either a stimulatory or inhibitory impact on the growth kinetics of *L. monocytogenes* during formation of the dual species biofilms on the SS surface (Figure 1a). While *S. baltica* significantly (p < 0.05) increased the μ_{max} (0.59 h^{-1}) of *L. monocytogenes* in dual species biofilms, it also significantly (p < 0.05) prolonged the lag time of *Listeria* to 22 h (Figure 1a, Table 1). In contrast, while *P. fluorescens* and *S. proteamaculans* significantly (p < 0.05) reduced μ_{max} (0.32 and 0.22 h^{-1}, respectively) of *L. monocytogenes* in co-culture with these Gram-negative food-spoilage bacteria, their presence significantly (p < 0.05) shortened the lag time to 12 h for *L. monocytogenes* in the joint biofilms (Figure 1a, Table 1). *L. monocytogenes* significantly (p < 0.05) reduced the μ_{max} to 0.32 and 0.27 h^{-1} for *P. fluorescens* and *S. proteamaculans*, respectively, while the growth rate (0.33 h^{-1}) was not significantly (p > 0.05) altered for *S. baltica* in the dual species biofilms (Figure 1b–d, Table 1). Moreover, the co-culture with *L. monocytogenes* introduced a significant (p < 0.05) lag phase of 4.8 and 12.4 h for *P. fluorescens* and *S. baltica*, respectively. However, in the case of *P. fluorescens*, the sampling frequency (every 12 h) made it difficult to verify the model estimate of λ = 4.8 h based on the experimental data (Figure 1b). The lag time remained close to 0 h for *S. proteamaculans* in co-culture with *Listeria* (Table 1, Figure 1c).

In terms of MPD, *P. fluorescens* and *S. proteamaculans* significantly (p < 0.05) lowered the N_{max} of *L. monocytogenes* in joint biofilms to $10^{5.57}$ and $10^{5.85}$ CFU cm^{-2}, respectively, while *S. baltica* significantly (p < 0.05) augmented the N_{max} ($10^{7.29}$ CFU cm^{-2}) reached by the pathogen (Figure 1a, Table 1). The impact of the co-culture biofilm with *L. monocytogenes* ranged from a significant (p < 0.05) reduction and increase in N_{max} for *S. baltica* and *P. fluores-*

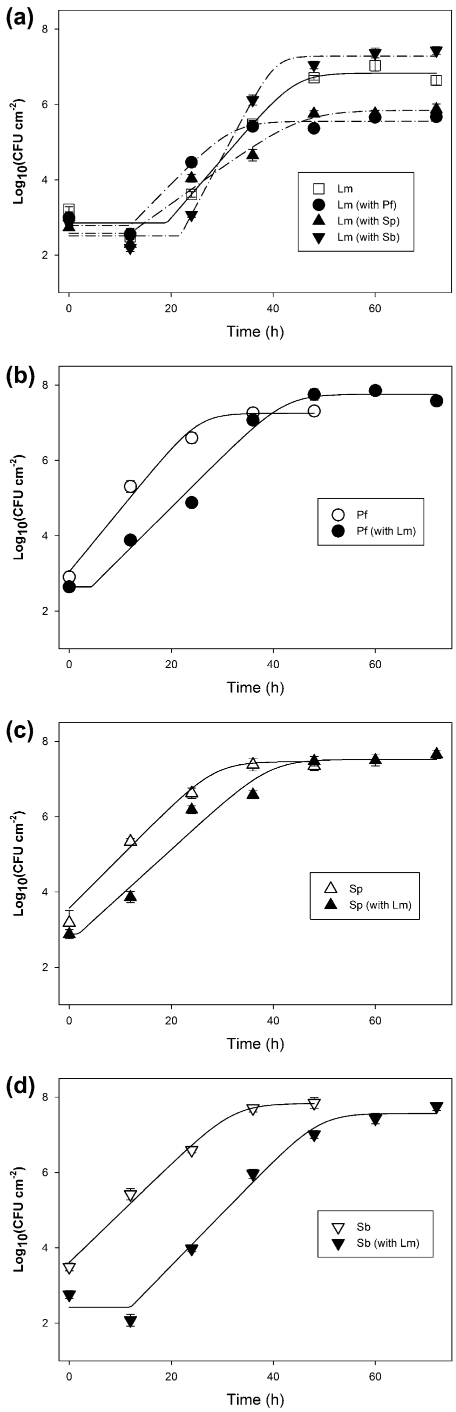

Figure 1. Growth of *L. monocytogenes*, *P. fluorescens*, *S. proteamaculans* and *S. baltica* in single or dual species biofilms on SS coupons (100% RH, 15 °C, 48 or 72 h). The microbial populations were enumerated on BHIA or selective agars (n = 6, ±SD). (a) *L. monocytogenes* single and dual species biofilms with each of the Gram-negative bacteria; (b) *P. fluorescens* single and dual species biofilm with *L. monocytogenes*; (c) *S. proteamaculans* in single and dual species biofilm with *L. monocytogenes* and (d) *S. baltica* in single and dual species biofilm with *L. monocytogenes*. Lines in the graph represent the numbers of bacteria predicted by the Logistic model with lag phase fits.

cens to $10^{7.56}$ and $10^{7.75}$ CFU cm^{-2}, respectively, to a negligible ($p > 0.05$) effect on N_{max} of *S. proteamaculans* (Figure 1b–d, Table 1).

Desiccation survival of L. monocytogenes *and the Gram-negative bacteria in single and dual species biofilms on SS coupons*

The formation of bacterial biofilm was followed by a three-week period of desiccation (43% RH and 15 °C), leading to overall losses in viability ($\Delta Log_{10}(N\ N_0^{-1})$) that ranged from 2.11 to 2.22 Log_{10}(CFU cm^{-2}) for *L. monocytogenes* and *S. proteamaculans*, respectively, to the significantly ($p < 0.05$) larger reductions of 5.24 to >8 Log_{10}(CFU cm^{-2}) for the populations of *P. fluorescens* and *S. baltica*, respectively (Figure 2a–d, Table 2). While the overall survival remained unchanged ($p > 0.05$) for *S. proteamaculans* and *S. baltica* biofilms together with *L. monocytogenes*, the survival of *L. monocytogenes* and *P. fluorescens* was significantly ($p < 0.05$) reduced in the dual species biofilms as compared to the single species biofilms (Table 2, Figure 2a–d). Populations of *S. baltica* dropped by *ca* 10^8 CFU cm^{-2} after 8–12 days (Figure 2d, Table 2) while residual levels of survivors were found after 21 days for the three other bacteria in both single and dual species biofilms (Figure 2a–c, Table 2).

Competition from the Gram-negative *P. fluorescens* and *S. proteamaculans* during the biofilm formation reduced the initial numbers of *L. monocytogenes* on the SS coupons, leading to significantly ($p < 0.05$) lower absolute survivor levels of 3.30 and 3.28 Log_{10}(CFU cm^{-2}), respectively, compared to 5.16 Log_{10}(CFU cm^{-2}) in the single *L. monocytogenes* biofilm (Table 2). *S. proteamaculans* outcompeted *L. monocytogenes* in the regrowth/biofilm formation achieving initial levels of >8 Log_{10}(CFU cm^{-2}) and thus constituted >99% of the surface population in joint biofilms. Due to its high desiccation resistance, final absolute survivor levels of 6.13–6.35 Log_{10}(CFU cm^{-2}) were significantly ($p < 0.05$) higher than for any of the other bacteria (Table 2).

To further analyse the survival kinetics of the bacteria in single and dual species biofilms the non-linear double-Weibull model was fitted to the normalized ($Log_{10}\ N\ N_{0-1}$) survivor curves. The α parameter [$Log_{10}(N_{01}\ N_{02-1})$] ranged 1.15–1.35 for *S. proteamaculans*, 1.94–2.84 for *L. monocytogenes*, 4.93–5.82 for *P. fluorescens* and 8.63–9.33 for *S. baltica* with no significant ($p > 0.05$) differences between α-values obtained for each bacterium in its single and dual species biofilms (Table 3). The lower α-values indicated presence of a higher proportion of a desiccation resistant subpopulation (N_{02}) in *S. proteamaculans* and *L. monocytogenes* than in *P. fluorescens* and *S. baltica*.

Table 1. Growth kinetics of *L. monocytogenes* and three Gram-negative bacteria during formation of single or dual species biofilms (100% RH, 15 °C, 48–72 h) in TSB-glu on SS coupons, as modelled using the logistic model with lag phase (Dalgaard 2009).

		Model estimates				Statistical indices		
Bacteria	Type of biofilm	N_{max}[I] (Log_{10} CFU cm^{-2})	μ_{max}[II] (h^{-1})	N_0[III] (Log_{10} CFU cm^{-2})	λ[IV] (h)	MSE_{model}	r^2	f
L. monocytogenes	Single	6.83^{d}[V] ± 0.08[VI]	$0.36^{b,c} \pm 0.03$	$2.85^{c} \pm 0.05$	$19.12^{b} \pm 0.85$	0.063	0.984	6.607 [VII]
	With *P. fluorescens*	$5.57^{f} \pm 0.02$	$0.32^{d} \pm 0.03$	$2.85^{c} \pm 0.08$	$12.00^{c} \pm 0.00$	0.037	0.980	2.832
	With *S. proteamaculans*	$5.85^{e} \pm 0.07$	$0.22^{f} \pm 0.02$	$2.58^{e,d} \pm 0.10$	$12.00^{c} \pm 0.00$	0.052	0.979	4.042
	With *S. baltica*	$7.29^{c} \pm 0.10$	$0.59^{a} \pm 0.03$	$2.52^{e,d} \pm 0.10$	$21.84^{a} \pm 0.36$	0.057	0.991	4.585
P. fluorescens	Single	$7.25^{c} \pm 0.06$	$0.38^{b} \pm 0.02$	$3.04^{b} \pm 0.06$	$0.00^{e} \pm 0.00$	0.049	0.984	3.151
	With *L. monocytogenes*	$7.75^{a} \pm 0.09$	$0.32^{d} \pm 0.01$	$2.62^{d} \pm 0.14$	$4.78^{d} \pm 0.62$	0.059	0.988	5.047
S. proteamaculans	Single	$7.46^{b} \pm 0.06$	$0.31^{d} \pm 0.01$	$3.57^{a} \pm 0.06$	$0.00^{e} \pm 0.00$	0.015	0.994	2.748
	With *L. monocytogenes*	$7.55^{b} \pm 0.07$	$0.27^{e} \pm 0.00$	$2.87^{c} \pm 0.09$	$0.12^{e} \pm 0.02$	0.104	0.973	6.783
S. baltica	Single	$7.84^{a} \pm 0.13$	$0.30^{e,d} \pm 0.01$	$3.61^{a} \pm 0.10$	$0.00^{e} \pm 0.00$	0.034	0.989	2.493
	With *L. monocytogenes*	$7.56^{b} \pm 0.08$	$0.33^{c,d} \pm 0.01$	$2.42^{e} \pm 0.04$	$12.37^{c} \pm 0.45$	0.072	0.988	5.090

Note: Bacterial counts were enumerated on BHIA or selective agars and converted into $Log_{10}(CFU\ cm^{-2})$ (n = 6).
[I] N_{max} denotes the maximum population density.
[II] μ_{max} represents the maximum specific growth rate.
[III] N_0 is the initial population level.
[IV] λ denotes the lag time.
[V] Values in the same column followed by same letters are not significantly different ($p > 0.05$).
[VI] Model estimate ± standard error of mean.
[VII] $f_{(26,\ 25)}$ table value for single *P. fluorescens*, single *S. proteamaculans* and single *S. baltica* is 1.92, whereas for all other biofilms the $f_{(38,\ 35)}$ table value is equal to 1.74.

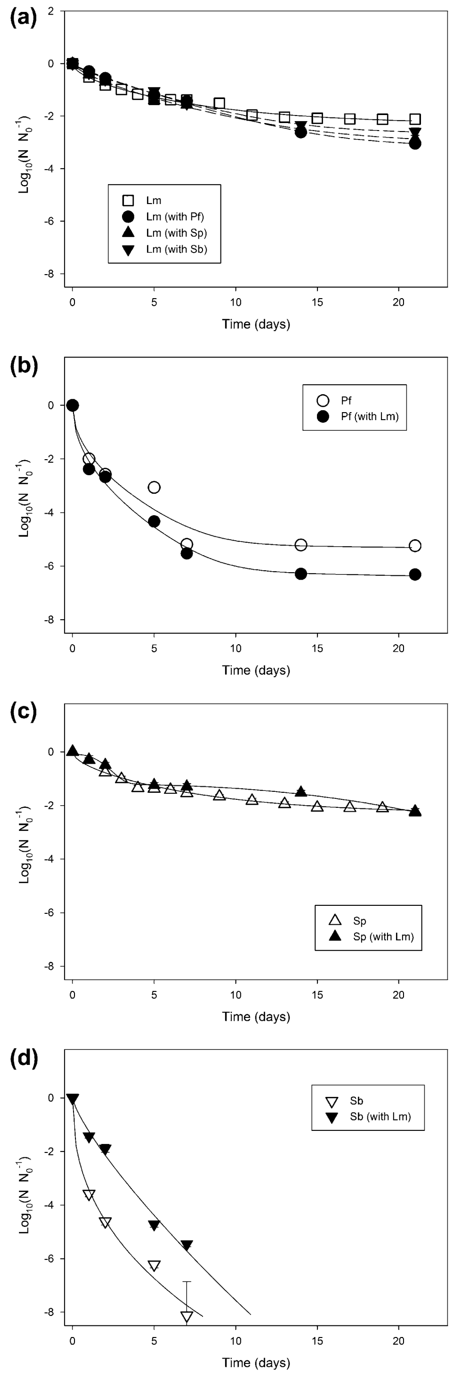

Figure 2. Desiccation survival of *L. monocytogenes*, *P. fluorescens*, *S. proteamaculans* and *S. baltica* on SS coupons in previously formed single and dual species biofilms (100% RH, 15 °C, 48 h) during exposure to 43% RH at 15 °C for 21 days. Survivors were enumerated on BHIA or selective agars (n = 6, ±SD). (a) *L. monocytogenes* single and dual species biofilms with each of the Gram-negative bacteria; (b) *P. fluorescens* single and dual species biofilm with *L. monocytogenes*; (c) *S. proteamaculans* in single and dual species biofilm with *L. monocytogenes* and (d) *S. baltica* in single and dual species biofilm with *L. monocytogenes*. Lines in the graph represent the numbers of survivors predicted by double-Weibull model fits. The detection limit is −8 $\mathrm{Log}_{10}(N N_0^{-1})$.

Table 2. Initial biofilm (48 h, 15 °C, 100% RH) and final desiccation survivor (21 day, 15 °C, 43% RH) populations in single and dual species biofilms formed by *L. monocytogenes*, *P. fluorescens*, *S. proteamaculans* and *S. baltica* on SS coupons.

Bacteria	Type of biofilm	Initial biofilm/regrowth population ($N_{0\ \mathrm{days}}$)[I] (Log_{10} CFU cm^{-2})	% in the dual species biofilm	Final desiccation survival population ($N_{21\ \mathrm{days}}$) (Log_{10} CFU cm^{-2})	Desiccation viability loss $\Delta\mathrm{Log}_{10}(N_{21}\ N_0^{-1})$ (Log_{10} CFU cm^{-2})	% of desiccation survivors in the dual species biofilm
L. monocytogenes	Single	7.27^{d}[II] ± 0.11[III]	–	$5.16^{b} \pm 0.14$	$-2.11^{a} \pm 0.17$	–
	With *P. fluorescens*	$6.33^{e} \pm 0.13$	0.48 ± 0.16	$3.30^{c} \pm 0.08$	$-3.04^{c} \pm 0.15$	89.53 ± 2.68
	With *S. proteamaculans*	$6.14^{e} \pm 0.11$	0.61 ± 0.25	$3.28^{c} \pm 0.14$	$-2.87^{c} \pm 0.18$	0.15 ± 0.05
	With *S. baltica*	$7.46^{d} \pm 0.11$	20.78 ± 7.32	$4.85^{b} \pm 0.14$	$-2.60^{b} \pm 0.18$	>99.99
P. fluorescens	Single	$8.49^{a} \pm 0.12$	–	$3.25^{c} \pm 0.10$	$-5.24^{d} \pm 0.16$	–
	With *L. monocytogenes*	$8.67^{a} \pm 0.10$	99.52 ± 0.16	$2.35^{d} \pm 0.12$	$-6.32^{e} \pm 0.15$	10.47 ± 2.68
S. proteamaculans	Single	$8.57^{a} \pm 0.09$	–	$6.35^{a} \pm 0.12$	$-2.22^{a} \pm 0.15$	–
	With *L. monocytogenes*	$8.39^{a} \pm 0.13$	99.39 ± 0.25	$6.13^{a} \pm 0.03$	$-2.26^{a} \pm 0.15$	99.85 ± 0.05
S. baltica	Single	$7.87^{c} \pm 0.06$	–	<DT[III]	$>8^{f}$	–
	With *L. monocytogenes*	$8.06^{b} \pm 0.11$	79.22 ± 7.32	<DT	$>8^{f}$	<0.01

[I] Counts (N_0 in desiccation experiment) include both biofilm and loosely attached cells as coupons which were not washed prior to desiccation.
[II] Numbers in the same column followed by different letters are significantly (p < 0.05) different. Mean ± standard deviation ($n - 1$).
[III] Below the detection limit of −0.4 $\mathrm{Log}_{10}(\mathrm{CFU\ cm}^{-2})$.

Table 3. Inactivation kinetics observed during desiccation (43% RH, 15 °C, 21 d) of single and dual species biofilms consisting of *L. monocytogenes* and three strains of Gram-negative spoilage bacteria on SS coupons.

		Model estimates[I]				Statistical indices		
Bacteria	Type of biofilm	α[II]	δ_1[III] (days)	p[IV]	δ_2[V] (days)	MSE_{model}	r^2	f[VII]
L. monocytogenes	Single	1.94^{c}[VI] ± 0.89	$3.06^{c,b} \pm 0.19$	$0.58^{c,d} \pm 0.19$	$266^{a} \pm 10$	0.017	0.960	1.418
	With *P. fluorescens*	$2.84^{c} \pm 1.39$	$4.28^{a} \pm 0.11$	$0.87^{c,b} \pm 0.17$	$134^{d} \pm 24$	0.013	0.989	1.069
	With *S. proteamaculans*	$2.32^{c} \pm 1.21$	$3.37^{b} \pm 0.14$	$0.73^{c,b,d} \pm 0.15$	$105^{d} \pm 12$	0.012	0.988	1.149
	With *S. baltica*	$2.24^{c} \pm 0.42$	$4.31^{a} \pm 0.09$	$0.84^{c,b} \pm 0.14$	$187^{c} \pm 27$	0.012	0.987	1.170
P. fluorescens	Single	$4.93^{b} \pm 0.48$	$0.35^{e,d} \pm 0.08$	$0.50^{c,d} \pm 0.22$	$233^{a,b} \pm 25$	0.196	0.945	23.724
	With *L. monocytogenes*	$5.82^{b} \pm 0.22$	$0.24^{e,d} \pm 0.04$	$0.49^{c,d} \pm 0.09$	$110^{d} \pm 8$	0.049	0.990	4.293
S. proteamaculans	Single	$1.35^{c} \pm 0.07$	$2.39^{c} \pm 0.07$	$1.04^{b} \pm 0.09$	$25^{e} \pm 0$	0.011	0.974	1.251
	With *L. monocytogenes*	$1.15^{c} \pm 0.05$	$2.63^{c} \pm 0.12$	$2.47^{a} \pm 0.20$	$21^{e} \pm 0$	0.012	0.979	1.401
S. baltica	Single	$8.63^{a} \pm 0.34$	$0.04^{e} \pm 0.00$	$0.38^{d} \pm 0.07$	$44^{e} \pm 10$	0.286	0.973	1.223
	With *L. monocytogenes*	$9.33^{a} \pm 0.14$	$0.78^{d} \pm 0.20$	$0.79^{c,b,d} \pm 0.08$	$206^{b,c} \pm 11$	0.053	0.996	7.955

Notes: Survivors were enumerated on BHIA or selective agars, converted into $Log_{10}(N N_0^{-1})$ (n = 6) and the double Weibull model (Coroller et al. 2006) and were then fitted to the transformed data.
[I]Model estimate ± standard error of mean.
[II]α denotes the proportion of subpopulation 1 to subpopulation 2.
[III]δ_1 represents the time to first decimal reduction in subpopulation 1.
[IV]P is the shape factor.
[V]δ_2 denotes the first decimal reduction time for subpopulation 2.
[VI]Values in the same column followed by same letters are not significantly different ($p > 0.05$).
[VII]The f-values were significant ($p < 0.05$) if $f < f$-table value. $f_{(86,\ 75)}$ table value for single *L. monocytogenes* and single *S. proteamaculans* is 1.52 and for all others $f_{(38,\ 35)}$ table is equal to 1.74.

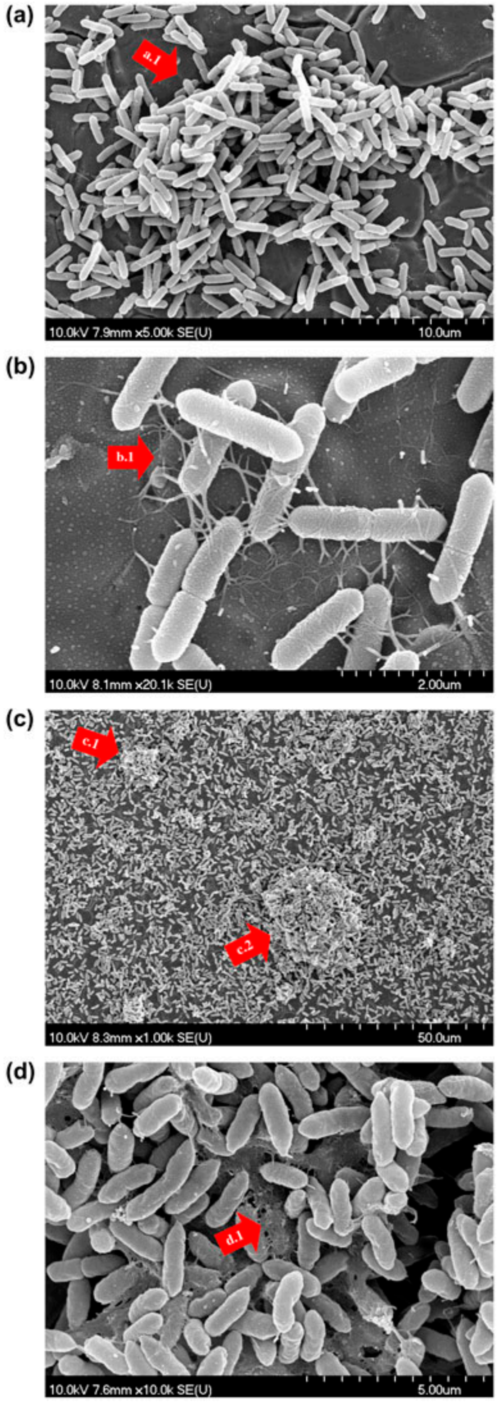

Figure 3. SEM images of *L. monocytogenes* and *P. fluorescens* in single species (a-b and c-d, respectively) and dual species biofilms (e and f) (varying magnifications, V_{acce}: 10 kV). The biofilms were developed on SS coupons (100% RH, 15 °C, 48 h) and post-fixed using modified fixation in cacodylate buffer (a, c, d and e), or the FC-72 solvent (b and f). The arrows a.1, b.1, c.1 and c.2 demonstrate the formation of microcolonies around niches and crevices on the surface. The arrow d.1 demonstrates the presumed EPS encasement of cells in the biofilm. The arrows e.1–4 indicate the formation and expansion of microcolonies in different directions. The arrow d.1 shows the presumed EPS encasement of cells in the biofilm as preserved by the organic fixation method.

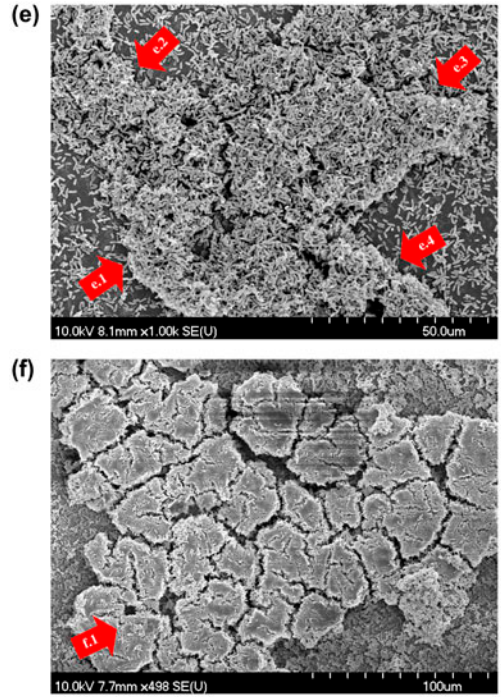

Figure 3. (*Continued*)

Bacterial δ_1- and δ_2-values, which represent the first decimal reduction time for subpopulations 1 and 2, respectively, were affected differently in the single and dual species biofilms being desiccated. The δ_1-value increased significantly ($p < 0.05$) for *L. monocytogenes* in co-cultured biofilms with *S. baltica* or *P. fluorescens* (δ_1 = ~4.3 days in both) compared to the single species biofilm (δ_1 = 3.06 days) and dual species biofilm with *S. proteamaculans* (δ_1 = 3.37 days) (Figure 2a, Table 3). A modest increase was seen for the δ_1-values of *S. baltica* which increased from 0.04 day in single species biofilm to 0.78 day in biofilms with *L. monocytogenes*. None of the other dual species biofilm δ_1-values differed significantly ($p > 0.05$) from those obtained in single species biofilms, indicating that the presence of *L. monocytogenes* had no effect on the initial desiccation survival of *S. proteamaculans* and *P. fluorescens* (Table 3). Mostly, bacteria in dual species biofilms exhibited significantly ($p < 0.05$) lower δ_2-values compared to values obtained in single species biofilms (Table 3). An exception was, where the presence of *L. monocytogenes* in the dual species biofilm increased the δ_2-value significantly ($p < 0.05$) for *S. baltica* as compared to the single species *S. baltica* biofilm. The values of shape parameter (p) ranged from 0.38 to 2.47 and were with the exception of *S. proteamaculans*, not significantly ($p > 0.05$) different for the individual bacteria when comparing their single and dual species biofilms (Table 3).

Microscopic evaluation of adhesion and biofilm formation

SEM showed that static biofilms formed by *L. monocytogenes* on SS coupons during incubation for 48 h at 15 °C consisted of microcolonies scattered across the surface with intermittent single cells (Figure 3a). Single cells can be seen to attach to the surface and other cells using flagella or fibril-like structures (Figure 3b). Neither the aqueous nor the organic-based solvent fixation method revealed formation of large amounts of EPS. Large microcolonies could be seen on the surface of SS coupons populated by *P. fluorescens* (Figure 3c), with close-up images showing how individual cells were connected by an amorphous mass, possibly EPS (Figure 3d). The dual species *L. monocytogenes* and *P. fluorescens* biofilm resembled the biofilm made by the Gram-negative bacterium alone (Figure 3e), which also constituted >99% of the population (Table 2). Images of the mixed biofilm fixed by the organic solvent showed cells embedded in an amorphous mass (Figure 3f). The cocci–bacilli-shaped *S. proteamaculans* formed microcolonies and expansively covered the SS surface (Figure 4a) with single cells being linked by a fibrous material (Figure 4b). The dual species biofilms were dominated by *S. proteamaculans* with a few interdispersed rod-shaped *L. monocytogenes* (Figure 4 c and d). When using the aqueous fixation method, SEM images of *S. baltica* revealed a thick multi-cellular layer completely covering the SS surface (Figure 5a) with no visible remnants of fibrous material, fibrils and flagella connecting the cells fixed (Figure 5b). The mixed biofilm resembled the single species biofilm made by *S. baltica* alone (Figure 5c). Interestingly, images obtained from the fixation method of FC-72 organic solvent fixation method showed cells in biofilms of *S. baltica* (dual species, Figure 5d) to be embedded in an amorphous gel-like material similar to that observed for *P. fluorescens*.

Epifluorescence microscope images obtained of dual species biofilms stained with the fluorescent Gram-stain kit revealed the formation of mono-special clusters within the biofilms seen as red (Gram-positive, *L. monocytogenes*) patches of varying sizes in a mostly green background consisting of the Gram-negative competitor bacteria (*P. fluorescens* and *S. proteamaculans*, Figure S1a and b, respectively, in the Supplementary material). The dual species biofilm with *S. baltica*, however, harboured a more equal distribution of the two bacterial species (Figure S1c).

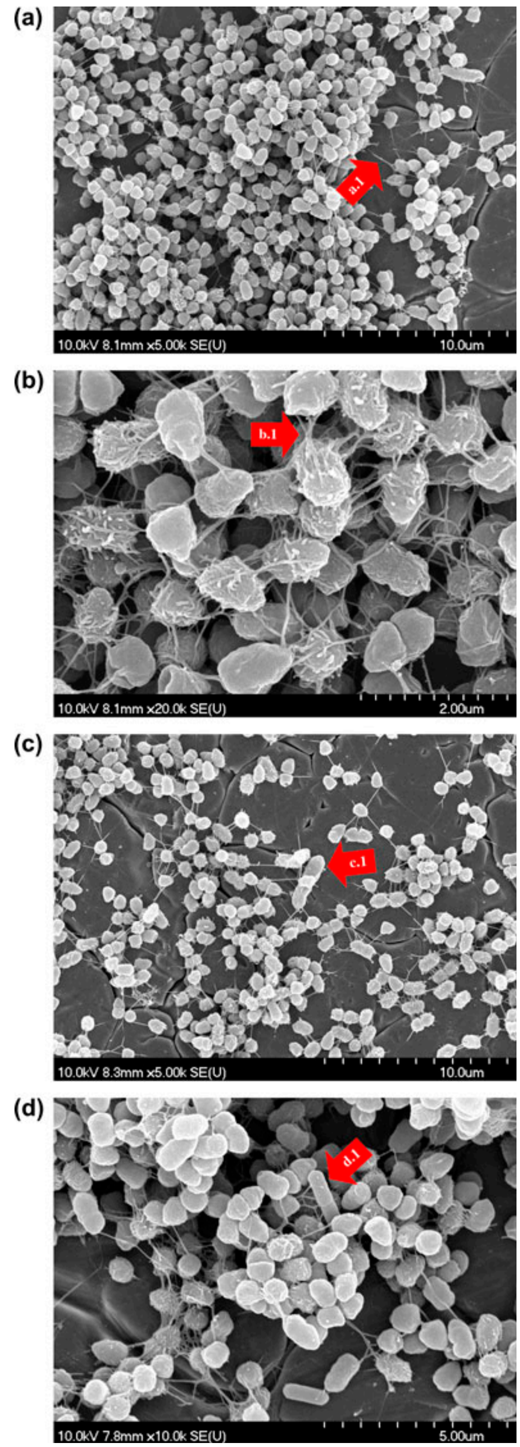

Figure 4. SEM images of *S. proteamaculans* in single (a and b) or dual species biofilms with *L. monocytogenes* (c and d) (varying magnifications, V_{acce}: 10 kV). The biofilms were developed on SS coupons (100% RH, 15 °C, 48 h) and post-fixed using modified fixation in cacodylate buffer. The arrows a.1 and b.1 demonstrate fibril-like junctures that appear to connect cells onto the surface and to each other at two different magnifications. The arrows c.1 and d.1 indicate, based on the difference in cell morphology, a *L. monocytogenes* cell adjacent to a *S. proteamaculans* microcolony.

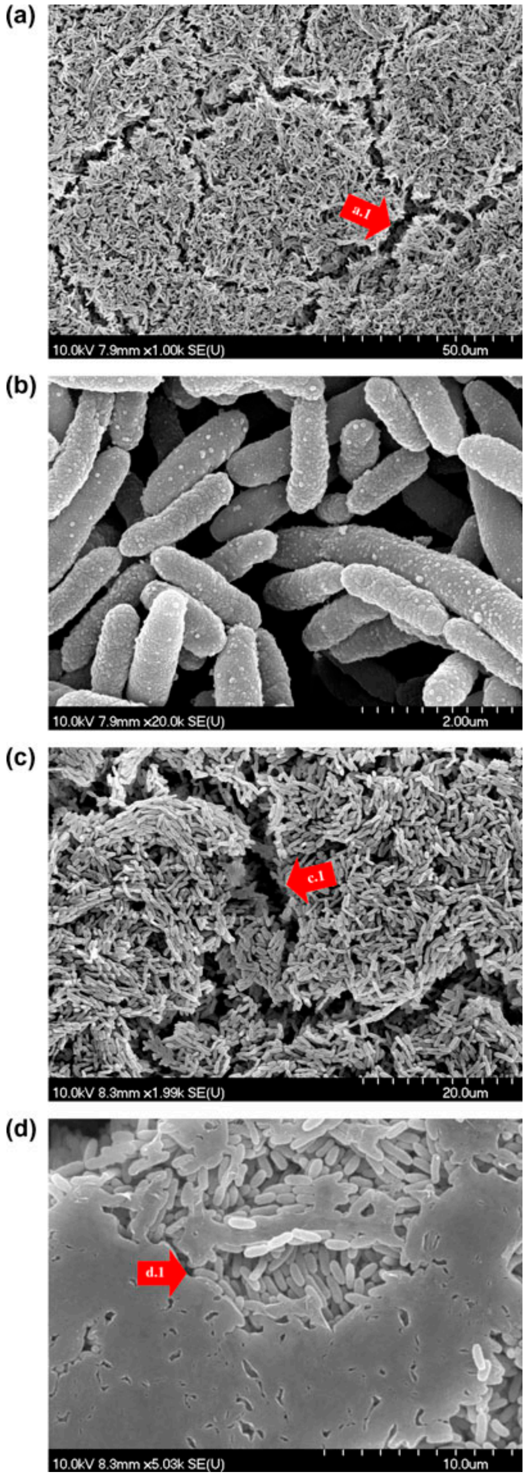

Figure 5. SEM images of *S. baltica* in single (a and b) or dual species biofilms with *L. monocytogenes* (c and d) (varying magnifications, V_{acce}: 10 kV). The biofilms were developed on SS coupons (100% RH, 15 °C, 48 h) and post-fixed using modified fixation in cacodylate buffer (a, b and c), or the FC-72 solvent (d). The arrows a.1 and c.1 show the microcolonies and cracks in the biofilm layer. The arrow d.1 indicates the same biofilm structure with cracks; however, cells are visibly encased in an amorphous EPS-like mass.

Discussion

The present study investigated the static regrowth/formation of biofilm of *L. monocytogenes* alone or in competition with common food-spoilage bacteria. The formation of biofilm was studied on the surface of food-grade SS at 15 °C to simulate what might take place in the food processing environment. To further characterize the interactions of bacteria in the dual species biofilms, the biofilm/regrowth kinetics was modelled using the Logistic model with lag phase (Dalgaard 2009). Within 48 h, initial contamination levels of 2.85 $Log_{10}(CFU\ cm^{-2})$ propagated to 6.83 $Log_{10}(CFU\ cm^{-2})$ (Table 1, Figure 1a) with SEM images confirming firm anchorage and establishment of microcolonies in the listerial biofilm (Figure 3a–b). Chavant et al. (2002) reported that the biofilms of *L. monocytogenes* on SS rose to densities of ~8 $Log_{10}(CFU\ cm^{-2})$ after two days at 20 °C with daily replenishment of the culture medium. The size of the population of *L. monocytogenes* biofilm is not easily compared among studies as it depends on factors such as the strain origin and serotype, carbohydrate production, the presence of nutrients and food soils, the pH and temperature (Chae et al. 2006; Folsom et al. 2006; Pan et al. 2010; Nilsson et al. 2011; van der Veen & Abee 2011).

The presence of Gram-negative spoilage bacteria at similar initial contamination levels affected the regrowth/formation of biofilm of the pathogen differently, with *S. baltica* stimulating its growth to final levels of 7.29 $Log_{10}(CFU\ cm^{-2})$ while the two other bacteria reduced listerial MPD numbers to 5.57–5.85 $Log_{10}(CFU\ cm^{-2})$ (Table 1, Figure 1a). All Gram-negative bacteria reached final populations of 7.25–7.84 $Log_{10}(CFU\ cm^{-2})$ in both single and dual species biofilms (Table 1, Figure 1b and c). *L. monocytogenes* introduced lag phases for *P. fluorescens* (4.78 h) and *S. baltica* (12.37 h) in the dual species biofilms, whereas no lag phase was observed in their respective single species biofilms (Figure 1 b and d). Autochthonous microbes in mixed species biofilms can have neutral, antagonistic and synergistic effects on the growth of *L. monocytogenes*. In a study of 29 food environmental isolates, Carpentier and Chassaing (2004) found that four strains stimulated growth of listerial biofilm, while 16 strains including *P. fluorescens* and *Serratia* sp. strains were inhibitory. At 15 °C *Staphylococcus sciuri* and *P. fluorescens* but not *Kocuria varians* were shown to decrease the formation of biofilm by *L. monocytogenes* on SS surfaces (Midelet et al. 2006). Guðbjörnsdóttir et al. (2005) also detected a decrease in biofilm formation by *L. monocytogenes* in dual species communities with *S. liquefaciens*, *Aeromonas* sp. and *P. fluorescens*. Also, various lactic acid bacteria have been shown to inhibit *Listeria* biofilms (Guerrieri et al. 2009; Winkelstroter et al. 2011; Zhao et al. 2013), however, inhibition depended on the nutritional environment (van der Veen & Abee 2011). In contrast, other studies reported that spoilage bacteria such as *Pseudomonas* (*P. putida* and *P. fragi*) and *Flavobacterium* spp. significantly increased biofilm formation by *L. monocytogenes* (Sasahara & Zottola 1993; Bremer et al. 2001; Hassan et al. 2004). Kostaki et al. (2012) found a neutral interaction between *Salmonella enterica* and *L. monocytogenes* growing in dual species biofilms at 15 °C. The interaction between *L. monocytogenes* and *S. baltica* in mixed species biofilms has not previously been investigated. However, the same *Shewanella* strain (A2) formed single and dual species biofilms with a *P. fluorescens* isolate (Bagge et al. 2001).

The Jameson Effect, which refers to the inhibition of growth in a batch culture of the competitor strains (here *L. monocytogenes*) by the dominant bacterium once its maximum cell concentration (N_{max}) has been reached (Ross et al. 2000), may explain the interactions observed between *L. monocytogenes* and the Gram-negative *P. fluorescens* and *S. proteamaculans*. This inhibition occurred after 36–48 for *P. fluorescens* and 48 h for *S. proteamaculans,* and was mainly caused by their significantly shorter lag phases (~0–5 h) compared to *L. monocytogenes* (~12–22 h, Table 1). Mellefont et al. (2008) showed how growth rates (μ_{max}) and initial contamination levels (N_0) impacted the Jameson Effect observed for *L. monocytogenes* in co-cultures with *E. coli*, *P. fluorescens* or *Lactobacillus plantarum*. Campo et al. (2001) found that *P. fluorescens* had no influence on the growth kinetics of *L. monocytogenes* at 10 °C, whereas four strains of Enterobacteriaceae significantly suppressed the N_{max} of *L. monocytogenes*. The lack of a Jameson Effect in co-cultures with *S. baltica* (A2) and *L. monocytogenes* at 4 and 27 °C had been described in an earlier study carried out in the authors' laboratory (Girard 2004).

The SEM images of the single species biofilms resembled those in other reports for static biofilms formed by *L. monocytogenes* (eg Chavant et al. 2002; Rieu et al. 2008), *P. fluorescens* (Simões et al. 2007), *Serratia* spp. (Teh et al. 2012) and *S. baltica* (Bagge et al. 2001). The anhydrous fixation method, which was originally developed to fix intestinal tract mucus (Allan-Wojtas et al. 1997), revealed the presence of an extracellular matrix in single and dual species biofilms with the Gram-negative bacteria. SEM images taken of the dual species biofilm with *L. monocytogenes* and *P. fluorescens* (Figure 3e and f) or *S. proteamaculans* (Figure 4c and d) resembled the Gram-negative single species biofilms in agreement with their quantitative dominance in the biofilms (Table 2). A thick biofilm layer was observed in SEM pictures of *S. baltica* single and dual species biofilms with *L. monocytogenes* where the latter constituted about 20% of the population (Figure 5 a and d, Table 2). Almeida et al. (2011) similarly reported that dual species *E. coli* and *L. monocytogenes* biofilms resembled single species *E. coli* biofilms in spite of *L. monocytogenes* making up nearly 35% of the population, indicating that one of the partner organisms in a mixed biofilm provides the scaffolding for the biofilm.

Using the fluorescing 'Gram-stain', *P. fluorescens* (green areas) was observed to dominate with areas of *L. monocytogenes* sparsely interdispersed (Figure S1a, Supplementary material), indicating segregation of the two strains in the dual species biofilm. Almeida et al. (2011) demonstrated that in dual species biofilms, *E. coli* and *L. monocytogenes* created two defined separate layers, while S*almonella enterica* and *L. monocytogenes* remained blended. In the present study, *L. monocytogenes* and *S. baltica* also segregated (Figure 5c and d, Figure S1c) whereas *S. proteamaculans* and *L. monocytogenes* cells blended (Figure 4c and d, Figure S1b) in the dual species biofilms.

The presence of the Gram-negative spoilage bacteria in the dual species biofilms negatively affected the proportion of *L. monocytogenes* in the biofilm and 21-day desiccation survival ($\Delta \mathrm{Log}_{10}(N_{21}\ N_0^{-1})$) (Table 2, Figure 2a). However, due to the significantly ($p < 0.05$, δ_1-values, Table 3) lower desiccation survival of *P. fluorescens* and *S. baltica*, the actual numbers of surviving *L. monocytogenes* cells after 21 days became significantly ($p < 0.05$, $N_{21\ \mathrm{days}}$, Table 2) higher than the numbers for the Gram-negative spoilage bacteria. In contrast, *S. proteamaculans* not only outcompeted *L. monocytogenes* during the formation of biofilm but also exhibited a desiccation tolerance that matched that of *Listeria* ($p > 0.05$, Table 2, $\Delta \mathrm{Log}_{10}(N_{21}\ N_0^{-1})$, Table 3, α and δ_1-values for single species biofilms), causing this spoilage bacterium to survive in high numbers.

The survival of surface associated *L. monocytogenes* during exposure to desiccation depends on the presence of food soils including proteins, oils and salt as well as the RH and prior osmoadapation (Vogel et al. 2010; Takahashi et al. 2011; Hingston et al. 2013). Moreover, the desiccation resistance of *L. monocytogenes* cells lodged in mature biofilms is enhanced, presumably due to the EPS within the biofilm matrix (Truelstrup Hansen & Vogel 2011; Hingston et al. 2013). Although the molecular mechanisms of listerial desiccation survival have not been fully elucidated, compatible solutes (osmolytes) such as glycine betaine, carnitine, proline and trehalose (Dreux et al. 2008; Ells & Truelstrup Hansen 2011) are likely involved.

The rise in biofilm EPS observed in dual species *L. monocytogenes* biofilms with *P. fluorescens* and *S. baltica* increased the listerial δ_1-values significantly ($p < 0.05$, Table 3) indicating a protective effect of the Gram-negative EPS. In line with this, listerial survival was observed to be significantly increased during desiccation (75% RH and 15 °C) in the EPS-rich dual species *Flavobacterium* spp./*L. monocytogenes* biofilm as compared to the single species listerial biofilm (Bremer et al. 2001)

In spite of the extensive biofilm formation by *P. fluorescens,* which saw the bacterium becoming encased in EPS (Figure 3), its desiccation survival was poor in comparison with *L. monocytogenes* and *S. proteamaculans* (Tables 2 and 3). The biofilm matrix, which may consist of alginate, eDNA, proteins and other polymers, has been reported to protect *Pseudomonas* spp. during water stress (Schnider-Keel et al. 2001; Chang et al. 2007; Mann & Wozniak 2012). In terms of the possible molecular mechanisms involved in the pseudomonad response to desiccation, a recent study detected upregulation of alginate synthesis genes (including algU) as well as flagellar genes during desiccation of *Pseudomonas putida* (Gülez et al. 2012).

Little is known about the desiccation survival of *S. proteamaculans* in food systems. Related Enterobacteriaceae, eg pathogenic *E. coli* and *Salmonella* spp., have been demonstrated to have high tolerance of dry conditions (Hiramatsu et al. 2005), while *Serratia liquefaciens* was shown to be less desiccation tolerant than *E. coli* (Berry et al. 2010). The strategies of desiccation survival for this group include production of extracellular cellulose, EPS, fimbriae, changes to membrane permeability and synthesis or uptake of compatible solutes such as trehalose (Ramos et al. 2001; White et al. 2006; Garmiri et al. 2008).

The desiccation survival of *S. baltica* (*S. putrefaciens*) has to the best of the authors' knowledge not previously been investigated. However, the bacterium has been shown to express a proteinaceous osmotic shock response that cross-protects during subsequent salt stress (Leblanc et al. 2003). *Shewanella* spp. are known to produce biofilms with considerable amounts of EPS (Figure 5, Bagge et al. 2001, Neal et al. 2007). However, similar to *P. fluorescens,* the bacterium exhibited a very low tolerance to the desiccation stress applied in the current study (Figure 2d, Tables 2 and 3). Consequently, the fact that the strongest biofilm and EPS formers, *P. fluorescens* and *S. baltica*, also were the most desiccation sensitive bacteria, points to EPS and/or the formation of biofilm not being good predictors of desiccation survival.

In conclusion, the presence of the Gram-negative food-spoilage bacteria reduced the listerial MPD during the formation of biofilm on SS coupons (100% RH, 15 °C, 48 h) and subsequently increased the desiccation viability loss of *L. monocytogenes* after exposure to dry conditions for 21 days at 43% RH and 15 °C. However, due to the higher desiccation resistance of *Listeria* in comparison to *P. fluorescens* and *S. baltica*, the final survivor population of the pathogen on day 21 was greater than those for the Gram-negative strains. In contrast, *S. proteamaculans* first outcompeted the pathogen during the formation of biofilm followed by an equal desiccation survival, leaving the Enterobacteriaceae to survive in *ca* 3 $Log_{10}(CFU\ cm^{-2})$ higher numbers than *L. monocytogenes* in the dual species biofilm. Therefore, under these simulated food plant conditions, the fate of *L. monocytogenes* during formation of dual species biofilms and desiccation depended mostly on the implicit properties of the co-cultured bacterium. Extending the results from this study, repeated rounds of uncontrolled formation of biofilm/desiccation cycles in the food processing plant would likely lead to domination by *L. monocytogenes* in an environment initially co-contaminated with *P. fluorescens* and *S. baltica*, while *S. proteamaculans* would outcompete *L. monocytogenes*.

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